

## C Programming Assignment

Topic: Function Declaration, Function Definition and Function Calling

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**COURSE AND BRANCH : B.tech AND CSE**

**FIRST YEAR**

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Q1. Identify the Function Declaration, Function Definition and Function Call.

```
#include<stdio.h>

void display();

int main()
{
    display();
    return 0;
}

void display()
{
    printf("Welcome");
}
```

Function Declaration: **void display();**

Function Definition: **void display()**

```
{  
    printf("Welcome");  
}
```

Function Call: **display();**

## Q2. Identify the Function Declaration, Function Definition and Function Call.

```
#include<stdio.h>
```

```
int add(int, int);
```

```
int main()
```

```
{  
    printf("%d", add(20,30));  
    return 0;
```

```
}
```

```
int add(int a, int b)
```

```
{  
    return a+b;
```

```
}
```

Function Declaration: **int add(int, int);**

Function Definition: **int add(int a, int b)**

```
{  
    return a+b;  
}
```

Function Call: **printf("%d", add(20,30));**

### Q3. Identify the Function Declaration, Function Definition and Function Call.

```
#include<stdio.h>

float area(float, float);

int main()
{
    float ans;

    ans = area(12,5);

    printf("%.2f", ans);

    return 0;
}

float area(float l, float b)
{
    return l*b;
}
```

Function Declaration: **float area(float, float);**

Function Definition: **float area(float l, float b)**

```
{
    return l*b;
}
```

Function Call: ans = **area(12,5);**

### Complete the Missing Part

### Q4. Complete the missing Function Declaration.

```
#include<stdio.h>
```

```
void message();
```

```
int main()
```

```
{
```

```
    message();
```

```
    return 0;
```

```
}
```

```
void message()
```

```
{
```

```
    printf("Hello Students");
```

```
}
```

Q5. Complete the missing Function Call.

```
#include<stdio.h>
```

```
void welcome();
```

```
int main()
```

```
{
```

```
welcome();
```

```
    return 0;
```

```
}
```

```
void welcome()
```

```
{
```

```
    printf("Welcome");
```

```
}
```

Q6. Complete the missing Function Definition.

```
#include<stdio.h>

void show();

int main()
{
    show();
    return 0;
}

void show()
{
    printf("C Programming");
}
```

**Q7. Complete the missing Function Declaration.**

```
#include<stdio.h>

int square(int n);

int main()
{
    printf("%d", square(5));
    return 0;
}

int square(int n)
{
    return n*n;
}
```

**Q8. Complete the missing Function Call.**

```
#include<stdio.h>

void college();

int main()
{
    college();

    return 0;
}

void college()
{
    printf("ABC College");
}
```

## Predict the Output (Q9 to Q13)

Q9.

```
#include<stdio.h>

void display();

int main()
{
    display();

    return 0;
}

void display()
{
    printf("Hello");
}
```

**Output: Hello**

Q10.

```
#include<stdio.h>
```

```
int add(int,int);

int main()
{
    printf("%d", add(15,25));

    return 0;
}

int add(int a,int b)
{
    return a+b;
}
```

**Output:40**

**Q11.**

```
#include<stdio.h>
void show();
```

```
int main()
{
    show();
    show();
    return 0;
}
```

```
void show()
{
    printf("Hi ");
}
```

**Output: Hi Hi**

## Q12.

```
#include<stdio.h>

int cube(int);

int main()
{
    printf("%d", cube(3));

    return 0;
}

int cube(int n)
{
    return n*n*n;
}
```

**Output: 27**

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## Q13.

```
#include<stdio.h>

void line();

int main()
{
    line();

    printf("C Programming");

    return 0;
}

void line()
{
    printf("*****\n");
}
```



**Output: \*\*\*\*\***

## **C Programming**

### Find the Error

**Q14. Find the mistake and rewrite the correct statement.**

```
#include<stdio.h>
```

**\\ declaration is missing**

```
int main()
```

```
{
```

```
    display();
```

```
    return 0;
```

```
}
```

```
void display()
```

```
{
```

```
    printf("Hello");
```

```
}
```

**Correction:**

```
#include<stdio.h>
```

```
void display();
```

```
int main()
```

```
{
```

```
    display();
```

```
    return 0;
```

```
}
```

```
void display()
```

```
{  
    printf("Hello");  
}
```

### Q15. Find the error.

```
#include<stdio.h>  
  
void print();  
  
int main()  
{  
    print;\\wrong function call statement  
    return 0;  
}
```

```
void print()  
{  
    printf("Welcome");  
}
```

#### Correction

```
#include<stdio.h>  
  
void print();  
  
int main()  
{  
    print();  
    return 0;  
}  
  
void print()  
{
```

```
printf("Welcome");  
}
```

## Q16. Find the error.

```
#include<stdio.h>  
  
int sum(int,int);  
  
int main()  
{  
    printf("%d", sum(5,10));  
    return 0;  
}
```

```
void sum(int a,int b)//IT SHOULD BE INT NOT VOID  
{  
    printf("%d", a+b);//IT SHOULD RETURN NOT PRINT  
}
```

### Correction

```
#include<stdio.h>  
  
int sum(int,int);  
  
int main()  
{  
    printf("%d", sum(5,10));  
    return 0;  
}  
  
int sum(int a,int b)  
{
```

```
    return a+b;
}
```

Q17. Find the error.

```
#include<stdio.h>

void test()// semi colon missing

int main()
{
    test();
    return 0;
}

void test()
{
    printf("Testing");
}
```

Correction:

```
#include<stdio.h>

void test();

int main()
{
    test();
    return 0;
}

void test()
{
    printf("Testing");
}
```

## Q18. Find the error.

```
#include<stdio.h>

void hello();

int main()
{
    hello();
    return 0;
}

void hello// curly brackets missing
{
    printf("Hello");
}
```

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Correction:

```
#include<stdio.h>

void hello();

int main()
{
    hello();
    return 0;
}

void hello()
{
    printf("Hello");
}
```

## Logical Questions

Q19. Arrange the following in the correct order.

Function Call

Function Definition

main() Function

Function Declaration

Write the correct sequence:

Function Declaration

main() Function

Function Call

Function Definition

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Q20. Write a C program that contains:

One Function Declaration

One Function Definition

One Function Call

The function should print:

Welcome to C Programming:

```
#include <stdio.h>

void displayMessage();

int main() {
    displayMessage();
    return 0;
}

void displayMessage()
{
    printf("Welcome to C Programming\n");
}
```

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